

Covariant Formulation and Variational Derivation of Maxwell's Equations

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Context

The conceptual structure of this derivation follows material introduced in advanced coursework at the Universitat de Barcelona. All detailed calculations, formal development, index manipulations, and typesetting are the independent work of the author.

Introduction

In classical mechanics, the action integral is introduced, and via the variational method, the Euler-Lagrange equations are obtained for the motion of a system. This variational procedure can be generalized to Classical Field Theory to deduce the equations of the electromagnetic field. We adopt the Système International (SI) conventions ($\tilde{\kappa} = \varepsilon_0$) to directly recover the standard physical field equations. Let the Lagrangian density be:

$$\mathcal{L}_{\text{em}}(A_\mu, \partial_\nu A_\mu) = -\frac{\varepsilon_0}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{c} J^\mu A_\mu$$

1. The Maxwell Action and Inhomogeneous Equations

The action functional governing the electromagnetic field is $S[A_\mu] = \int d^4x \mathcal{L}_{\text{em}}$. To obtain the equations of motion, we apply the principle of stationary action, $\delta S = 0$. We define and evaluate the variation of the integrand $\delta \mathcal{L}_{\text{em}}$ explicitly:

$$\begin{aligned} \delta \mathcal{L}_{\text{em}} &= -\frac{\varepsilon_0}{4} F^{\mu\nu} \delta F_{\mu\nu} - \frac{\varepsilon_0}{4} F_{\mu\nu} \delta F^{\mu\nu} - \frac{1}{c} J^\mu \delta A_\mu = -\frac{\varepsilon_0}{2} F_{\mu\nu} \delta F^{\mu\nu} - \frac{1}{c} J_\nu \delta A^\nu \\ &= -\frac{\varepsilon_0}{2} F_{\mu\nu} (\partial^\mu \delta A^\nu - \partial^\nu \delta A^\mu) - \frac{1}{c} J_\nu \delta A^\nu = -\frac{\varepsilon_0}{2} (F_{\mu\nu} \partial^\mu \delta A^\nu - F_{\mu\nu} \partial^\nu \delta A^\mu) - \frac{1}{c} J_\nu \delta A^\nu \\ &= -\frac{\varepsilon_0}{2} (F_{\mu\nu} \partial^\mu \delta A^\nu + F_{\mu\nu} \partial^\mu \delta A^\nu) - \frac{1}{c} J_\nu \delta A^\nu = -\varepsilon_0 F_{\mu\nu} \partial^\mu \delta A^\nu - \frac{1}{c} J_\nu \delta A^\nu \end{aligned}$$

where dummy indices have been relabeled to extract a common factor, utilizing the antisymmetry of the Faraday tensor $F_{\mu\nu} = -F_{\nu\mu}$. The variation of the global action is then constructed as:

$$\begin{aligned} \delta S &= \int d^4x \left(-\varepsilon_0 F_{\mu\nu} \partial^\mu \delta A^\nu - \frac{1}{c} J_\nu \delta A^\nu \right) \\ &= \int d^4x \left(-\varepsilon_0 \partial^\mu (F_{\mu\nu} \delta A^\nu) + \varepsilon_0 \partial^\mu F_{\mu\nu} \delta A^\nu - \frac{1}{c} J_\nu \delta A^\nu \right) \\ &= -\varepsilon_0 \int_{\partial V} d\Sigma F_{\mu\nu} \delta A^\nu + \int d^4x \left(\varepsilon_0 \partial^\mu F_{\mu\nu} - \frac{1}{c} J_\nu \right) \delta A^\nu \end{aligned}$$

Applying the divergence theorem and assuming the fields vanish at infinity, the boundary term evaluated over ∂V is discarded. Imposing $\delta S = 0$ for arbitrary variations δA^ν requires the

integrand to vanish identically:

$$\varepsilon_0 \partial^\mu F_{\mu\nu} - \frac{1}{c} J_\nu = 0 \implies \partial^\mu F_{\mu\nu} = \frac{1}{c\varepsilon_0} J_\nu \implies \partial_\mu F^{\mu\nu} = \mu_0 c J^\nu$$

These represent the inhomogeneous Maxwell equations in tensorial form. Using the definition $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, we can re-express the equations of motion in purely quadrivectorial form as $\square A^\nu - \partial^\nu (\partial_\mu A^\mu) = \mu_0 c J^\nu$. Furthermore, the antisymmetry of $F^{\mu\nu}$ naturally implies the covariant conservation of the four-current:

$$\partial_\mu \partial_\nu F^{\mu\nu} = -\partial_\mu \partial_\nu F^{\nu\mu} = -\partial_\nu \partial_\mu F^{\nu\mu} = -\partial_\mu \partial_\nu F^{\mu\nu} \implies \partial_\mu \partial_\nu F^{\mu\nu} = 0 \implies \partial_\nu J^\nu = 0$$

2. The Bianchi Identity and Homogeneous Equations

The second pair of Maxwell's equations arises as a pure geometric consequence of the antisymmetry of $F_{\mu\nu}$ via the Bianchi identities. Given a covariant tensor of rank 3, $T_{\nu\rho\lambda}$, its totally antisymmetric part contains all permutations. Letting $T_{\nu\rho\lambda} = \partial_\nu F_{\rho\lambda}$, we set its antisymmetrization to zero. Since even permutations are positive and odd permutations are negative, we use the Levi-Civita tensor to execute the expansion explicitly:

$$\begin{aligned} \partial_{[\nu} F_{\rho\lambda]} &= \epsilon^{\mu\nu\rho\lambda} \partial_\nu F_{\rho\lambda} = \epsilon^{\mu\nu\rho\lambda} \partial_\nu (\partial_\rho A_\lambda - \partial_\lambda A_\rho) = \epsilon^{\mu\nu\rho\lambda} \partial_\nu \partial_\rho A_\lambda - \epsilon^{\mu\nu\rho\lambda} \partial_\nu \partial_\lambda A_\rho \\ &= \epsilon^{\mu\nu\rho\lambda} \partial_\nu \partial_\rho A_\lambda - \epsilon^{\mu\nu\lambda\rho} \partial_\nu \partial_\rho A_\lambda = \epsilon^{\mu\nu\rho\lambda} \partial_\nu \partial_\rho A_\lambda + \epsilon^{\mu\nu\rho\lambda} \partial_\nu \partial_\rho A_\lambda = 2\epsilon^{\mu\nu\rho\lambda} \partial_\nu \partial_\rho A_\lambda \\ &= -\epsilon^{\mu\rho\nu\lambda} \partial_\nu \partial_\rho A_\lambda - \epsilon^{\mu\nu\lambda\rho} \partial_\nu \partial_\rho A_\lambda = -\epsilon^{\mu\rho\nu\lambda} \partial_\nu \partial_\rho A_\lambda - \epsilon^{\mu\rho\nu\lambda} \partial_\nu \partial_\rho A_\lambda = -2\epsilon^{\mu\rho\nu\lambda} \partial_\nu \partial_\rho A_\lambda \end{aligned}$$

Equating the symmetric and antisymmetric permutations requires:

$$2\epsilon^{\mu\nu\rho\lambda} \partial_\nu \partial_\rho A_\lambda = -2\epsilon^{\mu\nu\rho\lambda} \partial_\nu \partial_\rho A_\lambda \implies 4\epsilon^{\mu\nu\rho\lambda} \partial_\nu \partial_\rho A_\lambda = 0 \implies \partial_{[\nu} F_{\rho\lambda]} = 0$$

This yields the homogeneous Maxwell equations. Defining the dual tensor as $F^{*\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$, this identity is equivalently written as the vanishing divergence of the dual tensor:

$$\partial_\mu F^{*\mu\nu} = \partial_\mu \left(\frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} \right) = -\frac{1}{2} \epsilon^{\nu\mu\rho\sigma} \partial_\mu F_{\rho\sigma} = -\frac{1}{2} \epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} = -\frac{1}{2} \partial_{[\nu} F_{\rho\sigma]} = 0$$

3. Lorentz Invariants of the Electromagnetic Field

We construct scalar quantities from the electromagnetic tensor that remain invariant under Lorentz transformations. Contracting the field tensor with itself:

$$\begin{aligned} F^{\mu\nu} F_{\mu\nu} &= F^{0\nu} F_{0\nu} + F^{i\nu} F_{i\nu} = [F^{0j} F_{0j}] + [F^{i0} F_{i0} + F^{ij} F_{ij}] \\ &= F^{0j} (-F^{0j}) + F^{i0} (-F^{i0}) + F^{ij} F_{ij} = -2F^{0j} F^{0j} + \eta_{il} \eta_{js} F^{ij} F^{ls} \\ &= -2F^{0j} F^{0j} + \delta_{il} \delta_{js} F^{ij} F^{ls} = -2F^{0j} F^{0j} + \frac{1}{2} (F^{ij} - F^{ji}) F^{ij} \\ &= -2F^{0j} F^{0j} + \frac{1}{2} (F^{ls} \delta_{il} \delta_{js} - F^{ls} \delta_{is} \delta_{jl}) F^{ij} = -2F^{0j} F^{0j} + \frac{1}{2} (\delta_{il} \delta_{js} - \delta_{is} \delta_{jl}) F^{ls} F^{ij} \\ &= -2F^{0j} F^{0j} + \frac{1}{2} \epsilon_{ijk} \epsilon_{kls} F^{ls} F^{ij} = -2E^i E^i + 2c^2 B^i B^i = -2(\mathbf{E}^2 - c^2 \mathbf{B}^2) \end{aligned}$$

Similarly, contracting the field tensor with its dual yields:

$$\begin{aligned}
F_{\mu\nu}^* F^{\mu\nu} &= \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} F^{\rho\sigma} F^{\mu\nu} = \frac{1}{2} \left(\epsilon_{0ijk} F^{0i} F^{jk} + \epsilon_{i0kl} F^{i0} F^{kl} + \epsilon_{ij0l} F^{ij} F^{0l} + \epsilon_{ijk0} F^{ij} F^{k0} \right) \\
&= \frac{1}{2} \left(\epsilon_{0ijk} F^{0i} F^{jk} + \epsilon_{0ikl} F^{0i} F^{kl} + \epsilon_{lj0i} F^{lj} F^{0i} - \epsilon_{kji0} F^{kj} F^{0i} \right) \\
&= \frac{1}{2} F^{0i} \left(\epsilon_{0ijk} F^{jk} + \epsilon_{0ijk} F^{jk} - \epsilon_{0ijk} F^{kj} - \epsilon_{0ijk} F^{kj} \right) = \frac{1}{2} F^{0i} \left(4\epsilon_{0ijk} F^{jk} \right) \\
&= -4F^{0i} \left(-\frac{1}{2} \epsilon_{ijk} F^{jk} \right) = -4E^i (cB^i) = -4c (\mathbf{E} \cdot \mathbf{B})
\end{aligned}$$

The physical Lagrangian density corresponds to the first scalar, which in SI units ($\tilde{\kappa} = \epsilon_0$) takes the form $\mathcal{L}_{\text{em}} = \frac{1}{2} \left(\epsilon_0 \mathbf{E}^2 - \frac{1}{\mu_0} \mathbf{B}^2 \right)$.

4. Reduction to 3D Vectorial Form

We project the covariant formulation into the standard 3D vectorial Maxwell equations. We define the field vector components as $-\frac{1}{2} \epsilon_{ijk} F^{jk} = cB^i$ and $F^{0i} = E^i$. Evaluating the Bianchi identity for the temporal case ($\mu = 0$):

$$\begin{aligned}
\epsilon^{0\nu\rho\lambda} \partial_\nu F_{\rho\lambda} &= \epsilon^{0000} \partial_0 F_{00} + \epsilon^{0ijk} \partial_i F_{jk} = \eta_{lj} \eta_{sk} \epsilon^{0ijk} \partial_i F^{ls} = 0 \\
\implies \partial_i \left(-\frac{1}{2} \epsilon_{lsk} F^{ls} \right) &= 0 \implies \nabla \cdot (c\mathbf{B}) = 0 \implies \nabla \cdot \mathbf{B} = 0
\end{aligned}$$

Evaluating the spatial case ($\mu = i$):

$$\begin{aligned}
\epsilon^{i\nu\rho\lambda} \partial_\nu F_{\rho\lambda} &= \epsilon^{i0kl} \partial_0 F_{kl} + \epsilon^{ij0l} \partial_j F_{0l} + \epsilon^{ijk0} \partial_j F_{k0} = \epsilon^{i0kl} \partial_0 F_{kl} + \epsilon^{ijl0} \partial_j F_{l0} + \epsilon^{ijk0} \partial_j F_{k0} \\
&= \epsilon^{i0kl} \partial_0 F_{kl} + 2\epsilon^{ijk0} \partial_j F_{k0} = -\epsilon^{0ikl} \partial_0 F_{kl} + 2\epsilon^{ijk0} \partial_j F_{k0} \\
&= -\epsilon^{ijk} \partial_0 F_{jk} + 2\epsilon^{ijk} \partial_j F_{k0} = \partial_0 \left(-\epsilon_{ijk} \eta_{jl} \eta_{ks} F^{ls} \right) + 2\epsilon_{ijk} \partial_j \eta_{kl} \eta_{0s} F^{ls} \\
&= \partial_0 \left(-\epsilon_{ijk} \delta_{jl} \delta_{ks} F^{ls} \right) + 2\epsilon_{ijk} \partial_j \delta_{kl} \delta_{0s} F^{ls} = \partial_0 \left(-\epsilon_{ijk} F^{jk} \right) - 2\epsilon_{ijk} \partial_j F^{k0} = 0 \\
\implies \partial_0 \left(-\frac{1}{2} \epsilon_{ijk} F^{jk} \right) &= \epsilon_{ijk} \partial_j F^{k0} \implies \frac{1}{c} \frac{\partial}{\partial t} (cB^i) = \epsilon^{ijk} \partial_j (-E^k) \\
\implies \frac{\partial \mathbf{B}}{\partial t} &= -\nabla \times \mathbf{E}
\end{aligned}$$

Evaluating the inhomogeneous equations $\partial_\mu F^{\mu\nu} = \mu_0 c J^\nu$ for the temporal case ($\nu = 0$):

$$\partial_\mu F^{\mu 0} = \partial_i F^{i0} = \mu_0 c J^0 = \mu_0 c (\rho c) = \frac{\rho}{\epsilon_0} \implies \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

And for the spatial case ($\nu = i$):

$$\begin{aligned}
\partial_\mu F^{\mu i} &= \partial_0 F^{0i} + \partial_j F^{ji} = \mu_0 c J^i \implies \frac{1}{c} \frac{\partial E^i}{\partial t} + \partial_j F^{ji} = \mu_0 c J^i \\
\partial_j F^{ji} &= -\partial_j F^{ij} = -\frac{1}{2} \partial_j \left(F^{ij} - F^{ji} \right) = -\frac{1}{2} \partial_j \left(F^{ls} \delta_{il} \delta_{js} - F^{ls} \delta_{is} \delta_{jl} \right)
\end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{2}\partial_j \left[(\delta_{il}\delta_{js} - \delta_{is}\delta_{jl}) F^{ls} \right] = -\frac{1}{2}\partial_j \left(\epsilon_{ijk}\epsilon_{kls}F^{ls} \right) \\
&= -c\epsilon_{ijk}\partial_j \left(-\frac{1}{2c}\epsilon_{kls}F^{ls} \right) = -c(\nabla \times \mathbf{B})^i \\
&\implies \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} - c\nabla \times \mathbf{B} = \mu_0 c \mathbf{j} \implies \nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}
\end{aligned}$$

The Complete Maxwell Equations

The full set of field equations governing classical electrodynamics, derived directly from the covariant action functional:

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} & \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\
\nabla \cdot \mathbf{B} &= 0 & \nabla \times \mathbf{B} &= \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}
\end{aligned}$$